Anchored Integer Refinement of Redistricting Plans: A Kernel-Aware Matheuristic and Its Benchmarks

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Abstract

We study the integer form of a diffusion energy for redistricting, a kernel-weighted Laplacian cut term plus a fidelity term that anchors each move to the current plan, solved as a convex mixed-integer program on a merged pair of districts inside a tempered recombination chain. The result is a matheuristic ensemble generator, not a sampler, and we evaluate it as one. On Iowa it reduces mean cut edges by 20 percent over tempered ReCom across ten seeds, but simulated annealing over ReCom proposals reaches the same best plans at a fraction of the cost, so its value is anchored, kernel-aware refinement near a reference plan. We quantify that niche in a least-change experiment from Iowa's enacted map, where it ties the best annealer at a ten-county budget and is the only deterministic method, and in a multi-kernel showcase where perimeter and density kernels produce measurably different plans. A continuous relaxation with Cheeger-sweep rounding and a spectral mixture are reported as null results. The theory of the proposal, its exact correction, and the resulting sampler appear in a companion paper.

1 Introduction

2 Background: plans and ReCom

2.1 Dual graph and k-district plans

Let $G = (V, E)$ be the dual graph of a state, with vertices representing geographic units (precincts, census blocks, counties) and edges encoding shared boundaries. Each vertex v carries a population $p_v \geq 0$ and may carry additional attributes (demographic vectors, shapefile geometry). A *k-district plan* is a partition $P = \{V_1, \ldots, V_k\}$ of V such that each induced subgraph $G[V_i]$ is connected (*contiguity*) and the populations satisfy

$$(1-\varepsilon)\bar{p} \;\leq\; p(V_i) \;\leq\; (1+\varepsilon)\bar{p}, \qquad \bar{p} := \frac{p(V)}{k}, \tag{1}$$

for a tolerance $\varepsilon \in [0, 1)$. We write $\mathcal{P}_k(G; \varepsilon)$ for the set of valid plans. The *cut edges* of a plan are

$$\partial P \;=\; \{(u, v) \in E : P(u) \neq P(v)\},$$

and we call $|\partial P|$ the *cut-edge count* of P. This quantity is the discrete analog of the perimeter of the district boundaries and serves as our primary compactness metric [10].

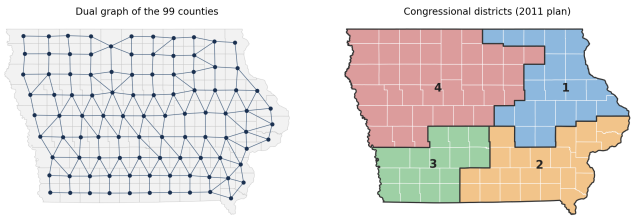


Figure 1: The dual graph of Iowa's 99 counties (left) and the four congressional districts enacted after the 2010 census (right). Nodes sit at county centroids; two counties are joined by an edge when they share a boundary segment.

2.2 Recombination (ReCom)

Recombination [8] is a Markov chain on $\mathcal{P}_k(G;\varepsilon)$ whose single step is given in Algorithm 1.

Algorithm 1: Vanilla ReCom step (Chain V)

Input: Current plan P, dual graph G, balance tolerance ε.

Output: Next plan P'.

Select two adjacent districts V_i, V_j uniformly at random from boundary pairs;

Form the induced subgraph $H = G[V_i \cup V_j]$;

repeat

  Sample a uniform random spanning tree T of H via Wilson's algorithm [14];

  Search edges of T for a cut e^* whose removal produces two ε-balanced components.

until *a cut e^* exists yielding two ε-balanced components*;

Define V_i', V_j' as the two components of $T \setminus \{e^*\}$;

$P' \leftarrow P$ with V_i, V_j replaced by V_i', V_j';

return P'.

The tree-based proposal flips many precincts simultaneously between V_i and V_j while preserving contiguity automatically (any edge removal from a tree produces two connected components). This is the key advantage over single-precinct flip chains, which traverse the state space too slowly to be practically useful.

Chains compared. We compare six chains, three of them from this work (Table 1).

Chain	Method	Mechanism	Stationarity claim
V	Vanilla ReCom [8]	spanning-tree split, always accept	unknown target
A	Tempered ReCom	ReCom + MH on cuts (symmetric-q)	approximate
B	ReCom + QP-diffusion mix (this work)	continuous QP, Cheeger sweep, repair	none (heuristic)
C	Tempered ReCom + IQP local (this work)	MIQP refinement, orphan-flip repair	none (matheuristic)
D	ReCom + spectral mix (after [7])	Fiedler + balanced sweep, no β, no repair	none (heuristic)
E	QP-MH (this work, Paper I, §4.6)	QP + Bernoulli rounding + exact MH	**exact**

Table 1: The six chains compared in this paper. The last column is the chain's relationship to the tempered target $\pi(P) \propto \exp(-\lambda|\partial P|)$. Only Chain E carries an exact Metropolis–Hastings correction; Chains B–D are heuristics whose output distributions we characterize empirically, and Chain A relies on the symmetric-q approximation (Paper I, §4.6).

Tempered ReCom uses a Metropolis–Hastings filter targeting $\pi(P) \propto \exp(-\lambda|\partial P|)$ with the symmetric-

q approximation (treating the forward and reverse proposal probabilities as equal); see Appendix A for the locked-in $\lambda = 0.10$. Our Chain D departs from published SpecReCom in one deliberate way. Instead of the sign split of the Fiedler vector (which ignores population balance), it runs our balance-constrained Cheeger sweep over the Fiedler order, closer in spirit to BalSpecReCom, so that all chains produce valid plans and the comparison isolates the role of the fidelity term. All chains also enforce the global two-sided balance window of Equation (3) on every move; we note (§4) that the raw GerryChain `recom` proposal alone does not.

The proposal we replace. The tree-and-cut step is kernel-blind (Paper I, §1.1), and the cut edge is chosen only after the tree is drawn, so the proposal cannot anticipate the shape of the split. The diffusion proposal of §3 replaces that step with a single convex optimization of a kernel-weighted Laplacian energy plus a locality-preserving fidelity term. The energy $x^\top L\, x = \sum_{(u,v)\in E} w_{uv}(x_u - x_v)^2$ equals the weighted cut count for binary x, so minimizing it directly minimizes the discrete perimeter in the chosen kernel.

3 The energy and its kernels

3.1 Locality kernels

Choosing the edge weights w_{ij} is the most important modeling decision. The kernel determines what *smoothness* means and therefore controls how district boundaries move under the relaxation. We restrict all weights to geographic neighbors,

$$w_{ij} = 0 \quad \text{if } (v_i, v_j) \notin E, \tag{2}$$

ensuring that diffusion operates only through geographic adjacency and preventing long-range interactions that could produce disconnected districts.

Uniform kernel (k_{unif}). $w_{ij} = 1$ for every edge. The Laplacian L is the combinatorial graph Laplacian; $x^\top Lx$ is exactly the cut edge count. In the $\beta \to 0^+$ limit the proposal approaches a balanced Fiedler-style split (§3.5).

Perimeter kernel (k_{perim}). $w_{ij} = |\partial v_i \cap \partial v_j|$ (shared-boundary length). The Laplacian quadratic form $x^\top Lx$ is then the *geographic perimeter* of the partition, weighted by boundary segment length. This kernel discourages fragmented districts and favors boundaries that cut along short segments.

Density kernel (k_{dens}). Let ρ_v denote the population density at vertex v (population per unit area). A Gaussian density kernel

$$w_{ij} \;=\; |\partial v_i \cap \partial v_j| \cdot \exp\!\Big(-\frac{(\rho_i - \rho_j)^2}{\sigma^2}\Big)$$

encourages cuts along sharp density gradients (urban–rural boundaries) while preserving the topological role of shared boundary length.

Demographic kernel (k_{demog}). Let $d_v \in \mathbb{R}^m$ be a vector of demographic attributes. A Gaussian similarity kernel $w_{ij} = \exp(-\|d_i - d_j\|^2/\sigma^2)$ encourages communities of interest. We define this kernel for completeness but do not use it in our experiments, and a caution stronger than "use as a soft regularizer" is warranted. If the attribute vector encodes race, a proposal that systematically shapes district boundaries around racial similarity risks making race the predominant districting criterion, which triggers strict scrutiny under the *Shaw v. Reno* line of racial-gerrymandering cases. Any practical use of demographic kernels should be reviewed against that doctrine, should document the kernel's weight relative

to race-neutral criteria, and is more defensible for attributes that are legitimate districting factors in the jurisdiction (e.g., municipal membership, communities of interest as defined by state law).

Composite kernels. Linear combinations $w_{ij} = \sum_k a_k w_{ij}^{(k)}$ with $a_k \geq 0$ remain valid, since they preserve nonnegativity and locality.

3.2 The convex QP

Let $H = G[V_i \cup V_j]$ be the merged subgraph of two adjacent districts. Let L_H be the kernel-weighted Laplacian of H, let $x_0 \in \{0,1\}^{|V_H|}$ encode the current bipartition (1 for vertices in V_i, 0 for vertices in V_j), and let p be the population vector restricted to H. Fix weights $\alpha, \beta > 0$ and balance tolerance $\varepsilon \in (0,1)$. Crucially, the balance constraint is anchored to the *global* per-district ideal $\bar{p} = p(V)/k$ from Equation (1), not to $p(V_H)/2$. Both sides of the split must individually satisfy the plan-validity bound, which yields the two-sided window

$$\ell \;=\; \max\big((1-\varepsilon)\bar{p},\; p(V_H) - (1+\varepsilon)\bar{p}\big), \qquad u \;=\; \min\big((1+\varepsilon)\bar{p},\; p(V_H) - (1-\varepsilon)\bar{p}\big). \tag{3}$$

Anchoring to $p(V_H)/2$ instead (a tempting simplification, and one present in an earlier version of our own implementation) only bounds the local split. Because $p(V_H)$ itself drifts as the chain moves population between pairs, district populations can ratchet outside the global tolerance by a factor of up to $(1+\varepsilon)$ per move on heterogeneous graphs. Our experiments audit the global bound at every step (§4).

Definition 3.1 (Continuous QP-diffusion)**.** *The continuous QP-diffusion bipartition of H is*

$$x^\star \;=\; \arg\min_{x \in [0,1]^{|V_H|}} \; \alpha\, x^\top L_H\, x \;+\; \beta\, \|x - x_0\|_2^2 \tag{4}$$

$$\text{subject to } \; \ell \;\leq\; p^\top x \;\leq\; u, \tag{5}$$

with ℓ, u as in Equation (3).

The first term in (4) is the kernel-weighted Laplacian smoothness energy. For binary x it equals the weighted cut count $\sum_{(u,v)\in E(H)} w_{uv}\, \mathbf{1}[x_u \neq x_v]$, and the continuous relaxation $x \in [0,1]^{|V_H|}$ extends this to a smooth objective on the unit cube. The second term is the *fidelity term*, which penalizes deviation from the current bipartition and ensures that each proposed move is local. The balance constraint enforces ε-balance on the proposed split.

Proposition 3.1 (Convexity and well-posedness)**.** *The objective in (4) is strictly convex on $[0,1]^{|V_H|}$ (since $\beta > 0$ and L_H is positive semidefinite). The feasible region defined by (5) is a nonempty polytope whenever the current plan is valid (it then contains x_0). Hence the QP admits a unique minimizer $x^\star$, computable in polynomial time by any convex QP solver.*

Spectral interpretation. Projecting the interior optimality condition $(\alpha L_H + \beta I)x^\star = \beta x_0$ onto the eigenbasis $\{\phi_k\}$ of L_H with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ yields the closed-form frequency response

$$\hat{x}_k^\star \;=\; \frac{\beta}{\alpha\lambda_k + \beta}\, \hat{x}_{0,k}, \qquad \hat{y}_k := \phi_k^\top y. \tag{6}$$

The QP therefore acts as a low-pass spectral filter on the prior x_0, with cutoff controlled by the ratio α/β. Equivalently, with $\tau := \alpha/\beta$, the optimum satisfies $(x^\star - x_0)/\tau = -L_H x^\star$, the implicit Euler discretization of the graph heat equation $\dot{x} = -L_H x$. Inspecting the leading eigenvectors $\phi_2, \phi_3, \ldots$ provides a sanity check on kernel choice. If they look like noise rather than coherent geographic regions, the chosen kernel is probably not informative.

3.3 Cheeger-sweep rounding and orphan-flip repair

The continuous optimum $x^\star \in [0,1]^{|V_H|}$ must be rounded to a binary bipartition. We use a *Cheeger sweep* [6]. Sort the entries of $x^\star$ in descending order, and for each threshold $k = 1, \ldots, |V_H| - 1$ form the candidate set S_k of the top-k vertices. Among candidates that satisfy ε-balance, return the one minimizing the cut count $|\partial(S_k, V_H \setminus S_k)|$. Candidates with disconnected sides are passed through the orphan-flip repair below before evaluation.

Algorithm 2: Cheeger-sweep rounding with orphan-flip repair

Input: Continuous optimum $x^\star \in [0,1]^{|V_H|}$, population vector p, target $\bar{p}$, tolerance ε, merged subgraph H.

Output: Binary bipartition $S \subset V_H$ or `FAIL`.

Sort V_H by $x^\star$ descending; call the order $v_{(1)}, \ldots, v_{(|V_H|)}$;

$\mathcal{C} \leftarrow \emptyset$; `// candidates`

for $k = 1$ **to** $|V_H| - 1$ **do**

  $S_k \leftarrow \{v_{(1)}, \ldots, v_{(k)}\}$;

  if $p(S_k) \in [(1-\varepsilon)\bar{p}, (1+\varepsilon)\bar{p}]$ *and both sides connected* **then**

    Add S_k to $\mathcal{C}$ with cut $|\partial(S_k, V_H \setminus S_k)|$;

  else if $p(S_k)$ *in balance and sides have* > 1 *components* **then**

    $S'_k \leftarrow$ orphan-flip repair of S_k (Algorithm 3);

    if S'_k *satisfies balance and connectivity* **then**

      Add S'_k to $\mathcal{C}$;

    end

  end

end

if $\mathcal{C} = \emptyset$ **then return** `FAIL`;

return $\arg\min_{S \in \mathcal{C}} |\partial(S, V_H \setminus S)|$.

The sweep can return a candidate S_k whose induced subgraph $G[S_k]$ or $G[V_H \setminus S_k]$ is disconnected. A single boundary vertex on the wrong side of the threshold is enough. Discarding such candidates outright would inflate the rejection rate; instead we repair them. The *orphan-flip* routine identifies the disconnected components of each side, keeps the largest component, and flips every smaller (*orphan*) component to the opposite side. Population balance is then re-verified; if the repair preserves both balance and connectivity, the repaired candidate enters the Cheeger sweep's candidate pool. Algorithm 3 formalizes this procedure.

Algorithm 3: Orphan-flip repair

Input: Candidate bipartition $(S, V_H \setminus S)$.

Output: Repaired bipartition or `FAIL`.

Identify connected components of $G[S]$ and $G[V_H \setminus S]$;

for *each side* **do**

  Keep the largest connected component;

  Flip every smaller component to the opposite side;

end

Verify both final sides are connected and ε-balanced;

if *verification fails* **then return** `FAIL`;

return repaired $(S', V_H \setminus S')$.

Why repair matters. On heterogeneous-population graphs such as Iowa, the unrepaired Cheeger sweep produces near-feasible candidates whose few orphan vertices (often singletons at the boundary) can be reassigned without affecting the cut count. Without repair, post-rounding rejection rates on Iowa run at 5–15%; with repair, $< 3\%$. SpecReCom [7] does not use repair (and we keep it out of our faithful Chain D implementation); without the β fidelity term, repair tends to accept lower-quality cuts.

On sweep guarantees. We do not claim the Cheeger bound [6] for the rounded solution. The sweep is applied to the constrained proximal optimum of Definition 3.1, not to the Fiedler vector, and the balance restriction on admissible thresholds voids the standard argument, so the sweep is a rounding heuristic whose output we measure. It also selects the candidate minimizing the *unweighted* cut even when the energy uses a non-uniform kernel, a configuration choice we flag in §4.3.

3.4 Hybrid composition with ReCom (Chain B)

Chain B's single step is a random mixture. With probability $p_R = 0.5$, take a Vanilla ReCom step (Algorithm 1); otherwise take a QP-diffusion step (Definition 3.1 and Algorithm 2). We state plainly what Chain B is and is not. *Neither branch carries a Metropolis correction*, the ReCom branch always accepts (as in Vanilla ReCom), and the QP-diffusion branch applies the rounded, repaired split unconditionally. Chain B therefore has no stationarity claim beyond ergodicity; it is an exploratory heuristic whose empirical output distribution we characterize against enumerated ground truth in Paper I, §5.1. The exactly corrected sampler built from the same proposal is Chain E (Paper I, §4.6).

Proposition 3.2 (Ergodicity of the hybrid)**.** *If ReCom is ergodic on $\mathcal{P}_k(G;\varepsilon)$ (which holds under standard assumptions; see [8, 2]), then the mixture of ReCom and the QP-diffusion step with any $p_R \in (0,1)$ is also ergodic.*

Proof. Write the mixture kernel as $K = p_R K_{\mathrm{R}} + (1-p_R)K_{\mathrm{Q}}$. If $K_{\mathrm{R}}^T(P,P') > 0$ for some T, then $K^T(P,P') \geq p_R^T K_{\mathrm{R}}^T(P,P') > 0$, so irreducibility of ReCom implies irreducibility of the mixture. For aperiodicity, the ReCom branch already holds in place with positive probability. The current split of any selected pair is a balanced bipartition of the connected merged subgraph H, and it is produced whenever the sampled spanning tree consists of a spanning tree of each side joined by one crossing edge, an event of positive probability under the uniform tree measure. Hence $K_{\mathrm{R}}(P,P) > 0$, so $K(P,P) > 0$ for every P and the chain is aperiodic. □

3.5 The integer QP-diffusion (Chain C)

The integer counterpart of Definition 3.1 replaces $x \in [0,1]^{|V_H|}$ with $x \in \{0,1\}^{|V_H|}$ and solves the resulting mixed-integer convex quadratic program (MIQP) directly.

Definition and solver.

Definition 3.2 (Integer QP-diffusion)**.** *The IQP local refinement of H is*

$$x^\star = \arg\min_{x\in\{0,1\}^{|V_H|}} \ \alpha\, x^\top L_H\, x \ + \ \beta\, \|x - x_0\|_2^2 \tag{7}$$

$$\text{subject to } \ \ell \ \leq \ p^\top x \ \leq \ u, \tag{8}$$

with the same global-ideal balance window (ℓ, u) of Equation (3) as the continuous case.

The objective is convex quadratic and the variable is binary. Gurobi's MIQP solver handles this directly via branch-and-bound; in practice, a 5-second time limit per solve is rarely hit on Iowa-sized merged pairs ($\sim$ 30–60 precincts).

Subsumption of cut-min; relation to SpecReCom.

Proposition 3.3 (Special cases)**.** *Setting $\alpha = 1$, $\beta = 0$, $w_{ij} \equiv 1$ in Definition 3.2 gives the classical minimum-cut integer program subject to the balance window. Hence Chain C subsumes the discrete cut-minimization MIP as a special case.*

We are careful *not* to claim the analogous statement for the continuous case. At $\beta = 0$ the relaxation of Definition 3.1 is degenerate. Any feasible constant vector $c\mathbf{1}$ attains energy zero, so the minimizer carries no cut information and the sweep is undefined. SpecReCom's Fiedler direction arises from the Rayleigh quotient *restricted to* $x \perp \mathbf{1}$, a constraint our proximal program does not impose; it is the fidelity term ($\beta > 0$) that breaks the constant degeneracy here, with x_0 playing the symmetry-breaking role that orthogonality plays in spectral partitioning. The precise relationship is that as $\beta \to 0^+$ with x_0 non-constant, the minimizer's projection off $\mathbf{1}$ aligns with the low-frequency eigenspace of L_H (Equation (6) with $\beta/(\alpha\lambda_k + \beta) \to 0$ fastest at high frequencies), so Chain B approaches a Fiedler-like split *in direction* while retaining balance feasibility. This gives a BalSpecReCom-style variant rather than SpecReCom itself. Chain D implements that variant directly.

Locality via the fidelity term. For $\beta = 0$, the IQP optimum is a minimum cut of H subject only to population balance and may differ arbitrarily from x_0. For $\beta > 0$ the optimum is biased toward x_0, since every flip incurs a β penalty, so the proposed split stays near the current one, is mostly contiguous already because x_0 is, and needs only a few boundary units repaired; the rejection figures of §3.3 are the measured consequence.

One step of Chain C. A single step of Chain C wraps an outer Tempered ReCom move (Algorithm 1 plus the Metropolis–Hastings filter of §3.4) around n_{iqp} deterministic IQP local-refinement passes. Each inner pass picks an adjacent district pair, solves the IQP on the merged subgraph, and applies orphan-flip repair; if the repair fails the inner pass leaves the plan unchanged. The full step is given as Algorithm 4.

Algorithm 4: One step of Chain C

Input: Current plan P, parameters $\alpha, \beta, n_{\text{iqp}}, \lambda, \varepsilon$.
Output: Next plan P'.
$P_0 \leftarrow$ Tempered ReCom step (Chain A) applied to P;
$P_{\text{curr}} \leftarrow P_0$;
for $t = 1$ **to** n_{iqp} **do**
  Select adjacent district pair (V_i, V_j) uniformly from boundary pairs of P_{curr};
  Form $H = G[V_i \cup V_j]$, build kernel-weighted L_H;
  Solve IQP (Definition 3.2) for $x^\star$;
  Apply orphan-flip repair (Algorithm 3);
  if *repaired bipartition feasible and connected* **then**
    Update V_i, V_j in P_{curr} to the new bipartition;
  else
    P_{curr} unchanged;
  end
end
return $P' \leftarrow P_{\text{curr}}$.

What Chain C is, and is not. The IQP loop is *deterministic* given the random pair choices. There is no Metropolis acceptance inside the loop, and the outer acceptance applies only to the Tempered ReCom move. We are explicit about the consequence. **Chain C is not a sampler for the tempered target** $\pi(P) \propto \exp(-\lambda|\partial P|)$, and no completion of the construction can make it one. A deterministic map T composed after a π-invariant kernel preserves π only if T is itself π-preserving (essentially a π-preserving involution [12]), whereas the IQP refinement is a many-to-one, cut-decreasing contraction onto local minima, the opposite of an involution. Nor can the refinement be folded into the proposal and Metropolized, since the reverse move would require the argmin at P' to output the strictly worse plan P, an event of probability zero. The distributional signature of this bias is visible in our own ground-truth data (Paper I, §5.1), where Chain C's cut marginal is shifted and sharply concentrated relative to Chain A's, despite the identical outer kernel.

We therefore present Chain C as what it is, a *matheuristic ensemble generator* in the lineage of ReCom-based optimization [4, 1], an iterated local search whose stochastic component (the tempered outer move) maintains diversity while the embedded MIQP drives compactness. This is a legitimate and, as §4 shows, empirically powerful tool for producing high-quality compact plans and for seeding or stress-testing ensembles. What it does not provide is a defensible null distribution for outlier analysis; for that use case the field's requirements are exactly the known-target property [3, 11] that Chain E (Paper I, §4.6) preserves and Chain C deliberately trades away for optimization power.

Trade-off: continuous vs. integer. The continuous relaxation is roughly 45× cheaper per step in wall-clock (a linear solve in the common case, by Paper I, Lemma 4.1), but on Iowa only the exact integer solve moves the compactness needle; the contrast is developed in §4.1.

4 Experiments

We evaluate the six chains of Table 1 on three datasets, namely the 5×5 grid (full enumeration, ground-truth target), the Iowa county dual graph (real geography, 99 counties, $k = 4$, $\varepsilon = 0.05$), and the North Carolina precinct dual graph (2 692 precincts, $k = 4$, $\varepsilon = 0.05$). All hyperparameters are locked in Appendix A; raw cut traces, per-step balance audits, and configuration stamps are released in the project repository.

Statistical protocol. Chains are autocorrelated, so we do not report i.i.d. test statistics on raw traces. For every comparison we report the mean difference, Cohen's d on the pooled post-burn-in standard deviation, and a z-test whose standard errors use effective sample sizes (ESS) from Geyer's initial-positive-sequence estimator of the integrated autocorrelation time. ESS values are reported alongside and are small, typically tens per 400 retained Iowa steps (as low as 3 for Chain E). On the 5×5 grid Chain C's trace is nearly constant and the estimator is unstable there, so we draw no single-run inference from it and let the ten-seed comparisons carry that claim. All ESS values are computed from cut traces and are appropriate for inference about cut means only; distributional tests use batch means (Paper I, §5.2). Each chain runs on its own seeded RNG stream. The tables below report a single seed (42) with full diagnostics; every headline comparison was replicated across ten seeds (42 through 51) under one solver configuration. On Iowa, Chain C's gain is -8.0 ± 0.4 cuts versus Chain A (-20.3%), negative on all ten seeds (paired $p = 6.4 \times 10^{-13}$), with C's mean cut count at 31.4 ± 0.3. Chain B's effect resolves to -0.9 ± 0.9 cuts (population standard deviation, as in every table) (-2.3%), negative on nine of ten seeds and nominally significant ($p = 0.015$) but an order of magnitude below C's, so single runs cannot distinguish it from zero. Chain D is indistinguishable from A ($+0.4$ cuts, $p = 0.10$). On the 5×5 fiber, C's between-seed spread is essentially nil (16.03 ± 0.02 mean cuts) and its -4.0% gain holds on all ten

seeds ($p = 4.3 \times 10^{-7}$), while B is null ($p = 0.59$). Between-seed standard deviations of the baselines are about 0.4 cuts on Iowa, so single-seed means are representative.

Balance audit. Every experiment records, at every step, the maximum relative deviation of any district population from the global ideal. All six chains satisfy the $\varepsilon = 0.05$ bound at every step of every run reported here (Iowa maxima 0.0494–0.0500 across the headline and ablation runs). We flag this because it is not automatic. GerryChain's raw `recom` proposal validates only one side of each tree cut, and an earlier version of our own local steps anchored balance to the merged-pair half rather than the global ideal (§3.2); either slip lets district populations drift to ∼4× the nominal tolerance on Iowa before the audit catches it. We recommend per-step balance auditing as standard practice for any custom chain harness that bypasses GerryChain's validator pipeline.

4.1 Compactness on Iowa: refinement gains and optimizer baselines

Chain	Mean cuts	Std	Δ vs A (Cohen's d)	ESS-z p	Wall-clock
V: Vanilla ReCom	41.39	4.78	—	—	1.2 s
A: Tempered ReCom	39.76	4.32	—	—	1.1 s
B: QP-diffusion mix	38.15	4.28	−4.0% (−0.37)	0.24	1.3 s
C: IQP local ($n_{\text{iqp}} = 10$)	**31.78**	**1.25**	−20.1% (−2.51)	$< 10^{-11}$	57.2 s
D: Spectral mix	39.51	4.70	−0.6% (−0.06)	0.85	3.9 s
E: QP-MH exact	55.21†	7.88	n/a (not converged)	n/a	3.6 s

Table 2: Iowa cut counts ($k = 4$, $\varepsilon = 0.05$, 500 steps, burn-in 100, seed 42, per-step balance audit passed by all chains). Chain C delivers a 20.1% reduction in mean cuts versus Tempered ReCom on this seed (−20.3% across ten seeds) with the tightest distribution; Chains B and D are statistically indistinguishable from A once balance is enforced correctly and autocorrelation is accounted for. †Chain E is not a competitor on compactness. It is the exact sampler for the tempered target, and its trajectory is still rising at 500 steps (ESS ≈ 3), so we report its mean as a *non-equilibrium lower bound* on the target's cut level, with no test statistic. Even as a lower bound it shows the target $\pi \propto \exp(-0.1\,|\partial P|)$ is far *less* compact than the output of any ReCom-family chain.

Iowa (6 chains). Three readings of Table 2. First, the integer refinement is the only local kernel whose Iowa gain survives rigorous accounting. A change of −20.1% on this seed (−20.3% across ten seeds) is a large difference within the chain family (the reported $d = -2.51$ scales it by the chain's equilibrium spread, which is not sampling noise, and should be read descriptively), though §4.1 shows classical annealing reaches the same best plans far more cheaply, and it is robust to the balance-constraint correction that erased Chain B's previously-measured advantage (with the global window enforced, B's point estimate swings between −0.3% and −4.0% across equivalent reruns and never approaches significance at these effective sample sizes; §6.1). Second, Chain D's parity with A and V supports our claim that the spectral cut alone, without the fidelity term, does not improve ReCom-family compactness on heterogeneous county graphs. Third, Chain E's trajectory shows the Iowa version of the calibration result of Paper I, §5.1. Taking E's non-equilibrium mean as a lower bound, every ReCom-based chain here sits at least ∼14–23 cuts *below* the tempered target's own mean, because the spanning-tree proposal is itself strongly compactness-biased. "Tempered ReCom targets π" is, on this evidence, a statement about the filter, not the samples.

The wall-clock column completes the practitioner picture. Chain C's compactness costs ∼ 47× the per-step time of Vanilla ReCom at $n_{\text{iqp}} = 10$ (about 0.11 s per step on Iowa), which is acceptable for ensembles of thousands of plans but material at precinct scale. Chain C's tight standard deviation (1.25

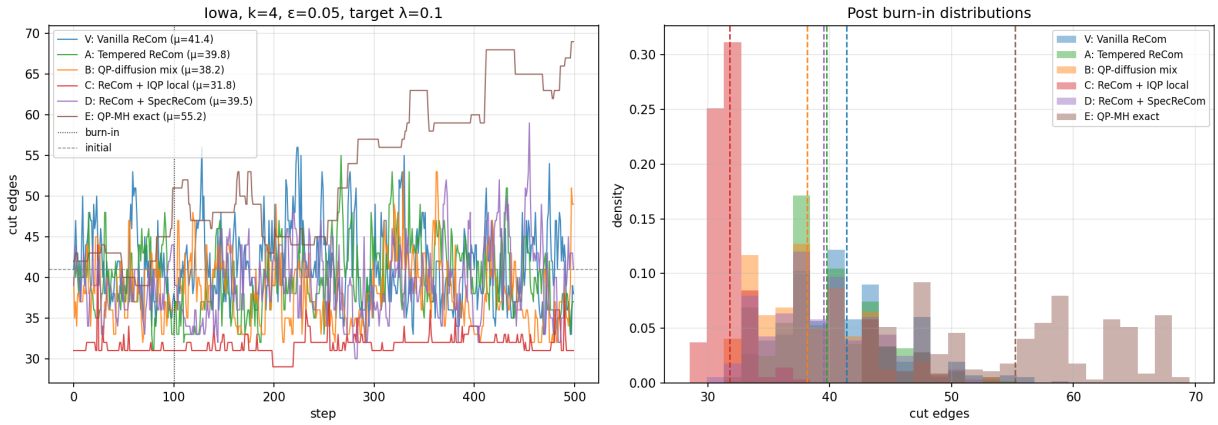


Figure 2: Cut-edge trace and post-burn-in histogram for the six chains on Iowa. Chain C concentrates tightly at low cut counts; Chain E drifts upward toward the tempered target's own cut level.

vs. 4.32 for A) is the signature of an optimizer concentrating on IQP local minima rather than the spread of a sampler at equilibrium, consistent with its matheuristic reading (§3.5).

Optimizer baselines. Because Chain C is benchmarked as an optimizer, we compare it against the classical optimizers a referee should demand. Simulated annealing over ReCom proposals (the same Metropolis filter as Chain A with a geometric cooling schedule, λ from 0.05 to 5) and pure hill climbing (accept only non-worsening moves) both cost the same as Vanilla ReCom, about one second for 500 steps on Iowa. We also run the same schedule stretched to 5 000 steps and a 25-restart variant.

Method	Best cuts	Ensemble mean	Distinct plans	Pairwise dist.	Wall
SA (500 steps)	31	31	1 (frozen)	—	1.0 s
Hill climbing (500)	31	31	1 (frozen)	—	0.9 s
SA (5 000 steps)	29	31	1 (frozen)	—	9.7 s
SA, 25 restarts	29	31.00	23	25.1	26 s
Vanilla ReCom + IQP	31	31.68	moving	—	53 s
Chain C	29	32.34	135	23.5	57 s

Table 3: Optimizer baselines on Iowa (seed 42, Gurobi 13). "Ensemble mean" is the mean cut count of the plans each method ends up producing (the frozen plateau for single-run SA and hill climbing, the set of 23 distinct final plans for restarted SA, the distinct post-burn-in plans for the chain methods). "Distinct plans" counts distinct plans produced (over 400 post-burn steps for Chain C). "Pairwise dist." is the mean number of counties reassigned between two distinct plans under the best district relabeling. The single-run SA and hill-climbing rows were replicated across ten seeds. SA's best ranges over 30–32 and never reaches 29 in 500 steps (mean frozen plateau 31.8), and hill climbing lands at 31 on every seed.

We report this comparison as a main finding rather than bury it, and it cuts both ways. The best plan quality found (29 cut edges) is reached by three routes, namely the 5 000-step anneal (9.7 s), restarted annealing (26 s), and Chain C (57 s). For a *single* compact plan, the longer anneal is the clear winner at about a sixth of Chain C's cost, so nobody should reach for an MIQP to find one compact map. As *ensemble* generators the two are closer. At under half of Chain C's wall-clock, restarted annealing yields a slightly better mean over its distinct plans (31.00 vs. 32.34) and slightly higher diversity (25.1 vs. 23.5 counties reassigned between plan pairs), while Chain C yields far more distinct near-optimal plans per

minute (about 141 vs. 53). Neither dominates. A 2×2 decomposition sharpens where the compactness comes from. Adding IQP refinement to Vanilla ReCom is worth about 10 cuts on the mean (41.39 to 31.68) and adding it to Tempered ReCom about 8 (39.76 to 31.78), while the tempering filter alone is worth about 1.6 cuts and moves the post-refinement mean by under 0.2. The optimizer does the work; the outer filter barely moves the mean, though on this seed only the tempered variant reached a 29-cut plan.

Two conclusions follow. On pure compactness optimization the classical baseline matches the MIQP machinery at a fraction of the cost, and we believe papers in the optimization-inside-ReCom line should report it as standard practice. What the energy retains is anchoring, small changes near a given plan, tested head to head in §4.4, together with kernel flexibility (§4.3) and, in Chain E, a provable stationary distribution that no annealing schedule provides.

4.2 North Carolina precinct validation

We repeat the six-chain comparison on the North Carolina precinct dual graph (`NC_VTD`, 2 692 precincts, $k = 4$, $\varepsilon = 0.05$, seed 42, 500 steps), roughly twenty-seven times the vertex count of the Iowa instance. This is a single-seed scale demonstration, so we report effect sizes without ESS-corrected test statistics and read the directions, not the exact magnitudes.

Chain	Mean cuts	Std	Δ vs A (Cohen's d)	Wall-clock
V: Vanilla ReCom	302.5	37.9	—	31 s
A: Tempered ReCom	226.4	23.8	—	30 s
B: QP-diffusion mix	252.6	50.9	+11.5% (+0.66)	89 s
C: IQP local ($n_{\text{iqp}} = 10$)	**145.7**	**3.9**	**−35.7% (−4.73)**	1950 s
D: Spectral mix	253.7	49.8	+12.0% (+0.70)	131 s
E: QP-MH exact	418.6†	25.1	n/a (not converged)	91 s

Table 4: North Carolina precincts ($k = 4$, $\varepsilon = 0.05$, 500 steps, burn-in 100, seed 42, per-step balance audit passed). Chain C reduces mean cut edges by 35.7% versus Tempered ReCom, a larger relative gain than on Iowa, at a per-step cost of about 3.9 s. †Chain E is still climbing at 500 steps and its mean is a non-equilibrium lower bound, as on Iowa.

Two things carry over from Iowa and one is new. The integer refinement again dominates its chain family, and by a wider margin (−35.7%, $d = -4.73$), consistent with our reading that heterogeneous real-world graphs leave more room for local optimization than the uniform grid. Chain C's distribution is again sharply concentrated (std 3.9 against 23.8 for A). The new observation is that on this larger graph the two mixture chains, B and D, land *above* Tempered ReCom rather than beside it (+11.5% and +12.0%), because a single continuous QP solve or spectral cut on a precinct-scale merged region is a coarse move that the unconditional accept step does not filter, so at scale the cheap regimes hurt. Only the exact integer solve (Chain C) and the exact sampler (Chain E, by construction) behave well at precinct scale. The cost is real. Chain C's 1950 s for 500 steps, about 3.9 s per step, is acceptable for a targeted refinement pass but not for a long ensemble run, which is the same speed-versus-power trade the Iowa timings show, magnified by scale.

4.3 Multi-kernel showcase on Iowa

We run Chain B and Chain C on Iowa with three kernels and visualize the resulting partitions. The three kernels are uniform (k_{unif}), perimeter (k_{perim}, w_{ij} = shared-boundary length from `shared_perim`),

and density (k_{dens}, $w_{ij} = |\partial v_i \cap \partial v_j| \cdot \exp(-(\rho_i - \rho_j)^2/\sigma^2)$ with $\rho_v = \text{TOTPOP/area}$ normalized to unit mean and $\sigma = 1$).

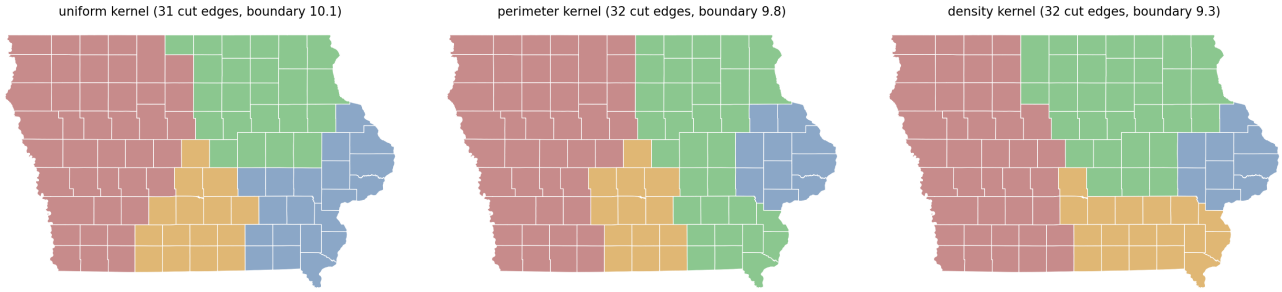


Figure 3: Final Chain C plans on Iowa under the three kernels (seed 42, 500 steps, identical constraints; only the Laplacian edge weights differ). Panel annotations give the unweighted cut count and the geographic boundary length (sum of shared-perimeter weights over cut edges). The plans differ in 19 to 25 of the 99 counties under optimal district relabeling. The kernel directly encodes geographic signal at the proposal level.

Kernel	B mean cuts	C mean cuts	C boundary (mean ± sd)	C $\lvert\Delta\rho\rvert$ ratio (mean ± sd)
k_{unif}	38.15	31.78	10.38 ± 0.67	1.78 ± 0.36
k_{perim}	39.00	33.39	$\mathbf{9.89 \pm 0.42}$	1.82 ± 0.39
k_{dens}	38.09	36.84	11.31 ± 1.16	$\mathbf{2.07 \pm 0.37}$

Table 5: Three kernels on Iowa ($k = 4$, $\varepsilon = 0.05$, 500 steps, burn-in 100, seed 42, corrected balance constraint, Gurobi 13). Mean cuts are post-burn-in averages of the unweighted cut count. The last two columns are averaged over every post-burn-in Chain C plan (not one snapshot), with the standard deviation across plans. "C boundary" is the geographic boundary length (sum of `shared_perim` over cut edges, in the dual graph's units). "$|\Delta\rho|$ ratio" is the mean population-density gap across cut edges divided by the graph-wide mean gap, so 1.00 means density-blind boundaries and larger values mean boundaries concentrated on urban–rural transitions. Each kernel is best on its own metric (bold): the perimeter kernel gives the shortest geographic boundary, the density kernel the highest density-gap ratio. Under non-uniform kernels the unweighted cut count is not the objective, so the cut columns must be read together with the kernel-specific columns.

What each kernel does to the plan. The three final Chain C plans in Figure 3 disagree on 19 to 25 of 99 counties under the best district relabeling, and averaged over the post-burn-in trajectory each kernel is best on the quantity it encodes (Table 5). The density kernel accepts a longer border in exchange for aligning cuts with population-density transitions; its four largest-gap cut edges are the borders of Polk County (Des Moines), then Linn (Cedar Rapids) and Johnson (Iowa City), the state's three urban cores, rather than edges through uniformly rural blocks.

We stress the trajectory averages over single-plan snapshots on purpose. A single final plan is a noisy draw, and its boundary and gap numbers can point the wrong way. Figure 3 itself illustrates this. The final density-kernel plan happens to land a shorter boundary than the final perimeter plan, opposite to the trajectory means. Only the post-burn-in means, with the spreads in Table 5, separate the kernel's systematic effect from run-to-run noise.

Reading the cut-count column. Chain B is nearly kernel-insensitive (38.1–39.0 mean cuts across all three), consistent with its Iowa null result. The continuous relaxation plus rounding does not transmit

the kernel's signal into the final integer plan the way the exact integer solve does.

4.4 Least-change refinement from the enacted plan

Remedial redistricting often asks for the smallest change that improves a specific plan, for example when a commission must repair a defect without redrawing the whole map. This is the setting the fidelity term was built for, and we quantify it directly. The reference plan is Iowa's enacted 2011 congressional map (47 cut edges; district populations within 0.01% of the ideal, so it is a valid state of every chain in this paper). A method may reassign at most D of the 99 counties, counted under the best district relabeling, and we record the best cut count it achieves within that budget, over ten seeds.

Three methods start at the reference. The *anchored IQP refinement* is the paper's inner move alone ($\alpha = 10$, uniform kernel, global balance window, no outer ReCom move), run for 50 passes. Because the fidelity weight β is exactly the least-change dial, the method's frontier is the envelope of (cuts, distance) trajectories over $\beta \in \{0, 0.3, 1, 3, 10, 30\}$, about 3 s of total solve time per seed. *Distance-capped annealing over ReCom proposals* uses the geometric schedule of §4.1 and rejects any proposal beyond the budget (2 000 proposals per budget, about 4 s). *Distance-capped annealing over single-county boundary flips*, with contiguity and the global balance window enforced per flip, is the fine-grained classical alternative (2 000 proposals, under 1 s).

Budget D	IQP anchored (β envelope)	SA-ReCom capped	SA-flip capped
3	42.40 ± 1.36	41.40 ± 0.80	$\mathbf{40.60 \pm 0.66}$
5	39.60 ± 0.92	38.80 ± 0.75	$\mathbf{38.20 \pm 0.40}$
10	$\mathbf{32.00 \pm 0.00}$	32.10 ± 0.30	33.10 ± 0.94
15	32.00 ± 0.00	$\mathbf{31.70 \pm 0.46}$	32.10 ± 0.30
20	32.00 ± 0.00	$\mathbf{31.20 \pm 0.40}$	31.60 ± 0.49
none	31.80 ± 0.40	$\mathbf{30.60 \pm 0.66}$	32.30 ± 0.46

Table 6: Least-change refinement of the enacted Iowa plan (47 cut edges). Entries are the best cut count reachable while reassigning at most D counties (best district relabeling), mean ± sd over ten seeds; bold marks the best method per budget. The last row removes the cap ($D = 99$).

Move granularity explains the pattern. From the reference, an accepted ReCom move reassigns a median of 26 counties (mean 25.0, minimum 1 over 197 accepted moves), so the capped ReCom annealer survives on the rare tree draws that happen to re-split almost identically. An accepted IQP pass at $\beta = 1$ moves 2.5 counties on average (median 2, maximum 5).

No method dominates the frontier, and we report all three regimes. At tight budgets ($D \leq 5$) capped flip annealing wins; one-county moves explore the small-perturbation space that both the tree re-split and the pair-optimal IQP step overshoot. At $D = 10$ the anchored refinement ties the capped ReCom annealer (32.00 against 32.10 ± 0.30, inside one standard deviation) and, unlike the annealers, it is deterministic. From the enacted plan the $\beta = 1$ refinement removes 15 of 47 cut edges while moving exactly 10 counties, reaching the same plan on all ten seeds in about half a second of solve time, with no cooling schedule or restart budget to tune. At loose budgets ($D \geq 15$) the task stops being least-change and capped ReCom annealing pulls ahead, consistent with §4.1. The practical summary is that for repairs moving roughly ten percent of units the anchored IQP matches the best annealer and is the only reproducible method, smaller repairs favor flip annealing, and larger redraws favor global annealing. An exact benchmark for this setting exists in the transfer-distance midpoint programs of Dobbs et al. [9], and comparing against it is the natural next step for this experiment. The IQP frontier here came from sweeping β, which confirms the fidelity term as a usable least-change dial rather than a fixed setting.

5 Ablations and sensitivity

All tables in this appendix are from the corrected balance constraint and Gurobi 13, on the same Chain C RNG stream as the headline table.

Configuration	Mean cuts	Std	Distinct-plan rate	Wall-clock
Baseline ($n_{\text{iqp}} = 5$)	31.64	2.10	32.8%	31 s
+ Jitter ($\mathcal{U}[1, 2]$ on edge weights)	32.40	1.93	**58.6%**	35 s
+ Tabu (size 3)	**30.16**	1.60	14.6%	30 s

Table 7: Diversity controls on Iowa ($n_{\text{iqp}} = 5$, $\alpha = 10$, $\beta = 1$). "Distinct-plan rate" is the fraction of the 500 steps that produced a not-yet-seen plan. The two controls pull in opposite directions, as intended.

Diversity controls (Jitter and Tabu). Jitter varies which cut is taken on a given pair and buys exploration, raising the distinct-plan rate from 33% to 59% at a cost of under one cut. Tabu varies which pair is picked and buys depth, forcing each step onto a boundary the chain has not just touched; it deepens the optimization to the best mean in the paper while halving the distinct-plan rate. A practitioner wanting a diverse ensemble turns on Jitter; one wanting the sharpest compactness turns on Tabu.

Configuration	Mean cuts	Δ vs A	Wall-clock
$n = 3, \alpha = 10, \beta = 1$	33.03	−16.9%	18 s
$n = 5, \alpha = 10, \beta = 1$	31.64	−20.4%	31 s
$n = 10, \alpha = 10, \beta = 1$	31.78	−20.1%	118 s
$n = 3, \alpha = 50, \beta = 1$	32.70	−17.7%	24 s
$n = 5, \alpha = 10, \beta = 0$	32.27	−18.8%	36 s
$n = 10, \alpha = 50, \beta = 0$	31.19	−21.6%	196 s

Table 8: Chain C parameter sensitivity on Iowa ($k = 4$, $\varepsilon = 0.05$, seed 42). n_{iqp} is the number of inner IQP passes per step. Wall-clock here wraps the whole run and was measured inside a 38-configuration sweep, whereas Table 2 sums per-step solve times on the same deterministic run, which is why the $n = 10$ row reads 118 s here and 57 s there. Compare rows within one table only.

Sensitivity to n_{iqp}, α, and β. Table 8 reports the uniform-kernel slice; the full sweep over $n_{\text{iqp}} \in \{3, 5, 10\}$, $\alpha \in \{10, 50\}$, $\beta \in \{0, 1\}$ and all three kernels (36 configurations) is released with the code as supplementary material. Three readings. First, n_{iqp} has diminishing returns. Moving from 3 to 5 passes buys about three and a half percentage points of cut reduction (−17% to −20%), but moving from 5 to 10 buys nothing more while multiplying the runtime by two to four depending on the timing convention (see the caption), because after a few passes the easy boundary improvements are exhausted and further passes revisit already-refined pairs. Five passes is the practical sweet spot. Second, α (the weight on the Laplacian smoothing term) matters little once it is large enough to dominate. Moving from 10 to 50 changes the result by half a cut or less, so we fix $\alpha = 10$. Third, the fidelity weight β helps rather than hurts. At $n = 5$, keeping $\beta = 1$ gives −20.4% against −18.8% with $\beta = 0$. The anchor keeps each refinement near the current plan, which on heterogeneous county populations lands better local optima than an unanchored solve, and it is what the exactly reversible Chain E needs to stay near-binary. We therefore keep $\beta = 1$ throughout.

6 Discussion

6.1 Limitations

(L1) Chain B adds nothing significant on Iowa (§4.1); the relaxation appears too loose on graphs with highly variable vertex populations.

(L2) Integer QP is computationally expensive. Chain C costs $\sim 47\times$ Vanilla ReCom per step at $n_{\text{iqp}} = 10$ on Iowa. For precinct-scale graphs, time limits with warm starts (and possibly coarsened merged-pair subproblems) will be necessary; the NC experiment (§4.2) is the pending scalability test.

6.2 Future directions

1. Benchmarking Chain C against exact cut-minimization optima on the public instances of Validi and Buchanan [13] and against the transfer-distance midpoint programs of Dobbs et al. [9].
2. Replica exchange or parallel tempering between Chain C and sampling chains, in the spirit of multiscale parallel tempering [5], to combine compactness with coverage.
3. A kernel-faithful Cheeger sweep that scores candidates by weighted cut (§3.3).

A Algorithm details and hyperparameters

A.1 Common setup

- Dual graph $G = (V, E)$ with vertex populations p_v and edge attributes (`shared_perim` for Iowa).
- k districts, ε-population balance per Equation (1), enforced as a two-sided bound around the global ideal on every move (Equation (3)) and audited per step. The bare GerryChain `recom` proposal is wrapped with an explicit whole-plan balance check (`recom_valid`), since its internal cut search validates only one side of each split.
- Initial partition from `gerrychain.tree.recursive_tree_part`, redrawn until all k districts satisfy the global tolerance (`balanced_tree_part`).
- Step count: 5×5: $N = 1\,000$, burn-in $B = 100$; Iowa: $N = 500$, $B = 100$; long-run validation (Paper I, §5.1): 2×10^5 (A) and 10^5 (E) steps.
- Random seeds: base seed 42 with a distinct offset per chain (independent streams for Python `random` and `numpy`).

A.2 Chain-specific parameters

A.3 Diversity controls

Weight jitter multiplies each edge weight by a draw from $\mathcal{U}[1, 2]$ before solving (always on for D, as in Davies et al.'s Algorithm 2; never used in Chain E, whose detailed-balance argument needs a deterministic QP map). Tabu forbids the last $K = 3$ pairs chosen by the local kernel, cleared whenever ReCom moves. Their effects are reported in Appendix 5.

Chain	Parameter	Value
A	λ (target temperature)	0.10
B	p_R (ReCom mix probability)	0.5
B	α (Laplacian weight)	10.0
B	β (fidelity weight)	1.0
B	Threshold scheme	Cheeger sweep, balance fallback
B	Laplacian normalization	False (unnormalized L)
C	n_{iqp} (IQP steps per outer iteration)	10
C	α	10.0
C	β	1.0
C	MIQP time limit	5.0 s
D	p_R	0.5
D	Weight randomization	$\mathcal{U}[1, 2]$ per edge
D	Repair	False; balanced sweep in place of sign split
E	α	1.0
E	β	10.0 ($\tau = \alpha/\beta = 0.1$; Paper I, Corollary 4.1 gives the resulting move size and P
E	Rounding	single-shot Bernoulli($x^\star$)
E	λ (MH target)	0.10

Table 9: Locked-in hyperparameters for the six chains.

A.4 Compute environment

- macOS (Darwin 23.6.0), Python 3.12.
- Gurobi 13.0.2 (academic license) for QP and MIQP, at its default thread count.
- NetworkX 3.x for graph operations.
- One Python process per run; no GPU. All experiments were run on a 2024 MacBook.

A.5 Diversity controls

Weight jitter multiplies each edge weight by a draw from $\mathcal{U}[1, 2]$ before solving (always on for D, as in Davies et al.'s Algorithm 2; never used in Chain E, whose detailed-balance argument needs a deterministic QP map). Tabu forbids the last $K = 3$ pairs chosen by the local kernel, cleared whenever ReCom moves. Their effects are reported in Appendix 5.

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